INCH-POUND

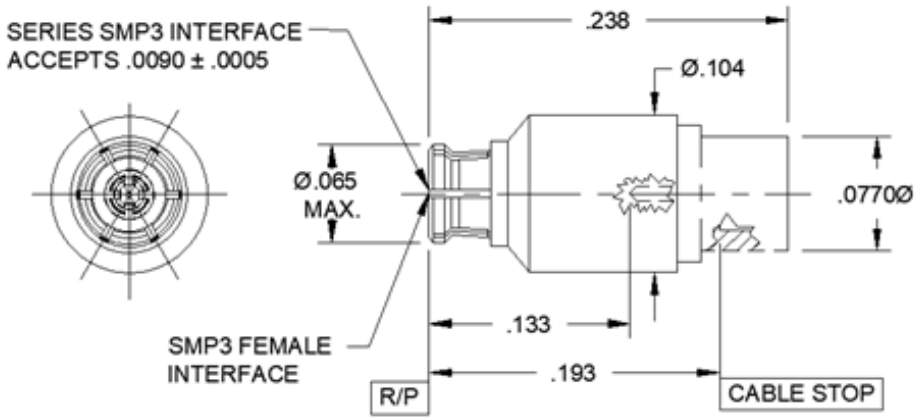
MIL-PRF-31031/33A
w/AMENDMENT 2
19 December 2025
SUPERSEDING
MIL-PRF-31031/33A
w/AMENDMENT 1
16 February 2024

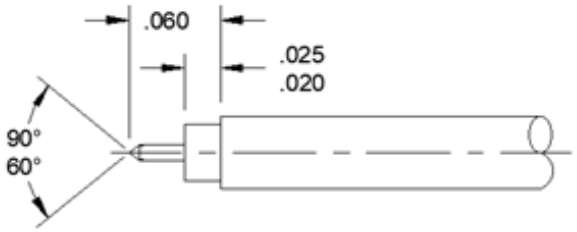
PERFORMANCE SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SOCKET CONTACT, SERIES SMP3, FOR .047 SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-31031.





PIN	Cable accommodations	Cable overall diameter
M31031/33	M17/151-00001 1/	.047 inch

FIGURE 1. General configuration.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5935

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NOTES:
1. Dimensions are in inches.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Special tools shall be required for assembly. Contact the manufacturer.
4. Interface in accordance with MIL-STD-348, series SMP3, socket contact.

FIGURE 1. General configuration, socket contact – Continued.

ENGINEERING DATA

Nominal impedance: 50 ohms

Frequency range: DC to 65 GHz.

RF leakage: -80 dB minimum to 3 GHz.

Voltage rating: 168 V rms maximum at sea level: 33 V rms at 70,000 feet.

Operating temperature: -55°C to +165°C.

REQUIREMENTS (after assembly):

Dimensions and configuration: See figure 1.

Mating interface: In accordance with MIL-STD-348.

Force to engage: 4.0 pounds maximum - full detent
2.0 pounds maximum - smooth bore

Force to disengage:

Full detent: 2.0 pounds minimum.

Smooth bore: 0.3 pounds minimum.

Radial misalignment: Not applicable.

Axial misalignment: Not applicable.

Center contact inspection conditions (after heat treat)

Oversize test pin:

Test pin diameter: .0098 inch +.0001/-.0000 inch.

Insertion depth: .025 inch minimum.

Number of insertions: 1.

Insertion force test:

Test pin diameter: .0095 inch +.0001/-.0000 inch.

Insertion force: 12 ounces, maximum.

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Insertion depth: .025 inch minimum.

Withdrawal force test:

Test pin diameter: .0085 inch +.0000/-.0001 inch.
Withdrawal force: 0.125 ounce, minimum.

Insertion depth: .025 inch minimum.

Coupling proof torque: Not applicable.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Center contact retention: 0.75 pounds minimum axial force.

Radial torque: Not applicable.

Voltage standing wave ratio:

DC to 12 gigahertz: 1.25:1 maximum.
12 to 26.5 gigahertz: 1.35:1 maximum.
26.5 to 40 gigahertz: 1.40:1 maximum.
40 to 65 gigahertz: 1.55:1 maximum.

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VSWR procedure.

The VSWR shall be measured in accordance with the following procedure or a method acceptable to the Government.

Part should be tested using a Network Analyzer with the Time Domain (TDR) option installed. This is essential to allow the effect of the adapters to be "gated" out. The recommended network analyzer systems include suitable equipment with NIST traceability or equivalent. All gating will be completed in accordance with the procedures stated by the network analyzer manufacturer with the following time domain settings:

TABLE I. Suggested Time Domain Settings.

Transform Mode 1/	Gate Type	Gate Shape 2/	Start Gate Time (ps)	Stop Gate Time (ps)
Low-pass	Bandpass	Minimum, Normal, Wide	End of Adapter 3/ – Rise Time of Gate 4/	End of Device Under Test + Rise Time of Gate

1/ The low-pass mode will have a higher resolution than the band-pass mode.
2/ Each shape specified will not be smaller than the minimum gate span to ensure the response has the correct start and stop times.
3/ "Port Extension" is also used for detection to move the measurement plane after calibration is complete.
4/ The minimum window is used for best resolution.
5/ The maximum window is used for best dynamic range.

The printer/plotter and the computer should be any unit compatible to the system. The test adapters will utilize the following frequency bands:

Frequency	VSWR
500 MHz	1.055
1 GHz	1.055
5 GHz	1.075
10 GHz	1.100
15 GHz	1.125
18 GHz	1.140
26 GHz	1.180
40 GHz	1.25
50 GHz	1.30
65 GHz	1.375

50-ohm precision air lines may also be used on both sides of the device under test.

Calibration of the system should be performed using the manufacturer's calibration kits and the recommended calibration procedures. The frequency range shall be DC to 65 GHz. The VSWR calibration test setup shall be verified using the manufacturers verification kits. The calibrated system VSWR shall be less than 1.02 + .001F (F in GHZ).

The VSWR of the DUT shall be measured using the procedures described in the manufacturer's operating instructions. The time domain shall then be used to remove the effects of the test adapter with the recommended settings as mentioned above.

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The output shall be generated using the appropriate printer/plotter.

Moisture resistance: MIL-STD-202-106, except step 7b shall be omitted. Resistance shall be 700 megohms within 5 minutes after removal from humidity.

Contact resistance (in milliohms, maximum):

	Initial
Center contact	6.0
Outer contact	2.0
Outer conductor to connector body	0.5

Dielectric withstanding voltage at sea level: 250 V rms minimum, 65 V rms at 70,000 feet.

Vibration, high frequency: MIL-STD-202-204, test condition D.

Corona level:

Altitude: 70,000 feet.
100 V rms, minimum.

Insulation resistance: 3,500 megohms minimum when tested in accordance with MIL-PRF-31031.

Altitude: Sea level to 70,000 feet.

Solderability: Applicable.

Shock (specified pulse): MIL-STD-202-213, test condition I.

Thermal shock: MIL-STD-202-107, test condition B (except high temperature to be +165°C).

RF high potential withstanding voltage:

150 V rms minimum at sea level.

Frequency: 5 MHz.
Leakage current: Not applicable.

Cable retention: 20 pounds minimum.

Coupling mechanism retention force: Not applicable.

Durability: 100 cycles.

RF insertion loss: $.12\sqrt{F(GH_z)}$ dB

Part or Identifying Number (PIN): M31031/33-0001 (See figure 1).

Materials:

Contacts and adapter housing: Beryllium copper in accordance with ASTM B196/B196M, gold plated to a minimum thickness of 50 microinches in accordance with MIL-DTL-45204, type II, grade C, class 1, over 50 microinches minimum of nickel in accordance with SAE-AMS-QQ-N-290.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-31031, this document references the following:

ASTM B196/B196M	MIL-STD-202-204
MIL-DTL-45204	MIL-STD-202-213
MIL-STD-202-106	MIL-STD-348
MIL-STD-202-107	SAE-AMS-QQ-N-290

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Preparing activity:
DLA - CC

(Project 5935-2026-010)

Review activities:
Amy - AT, AV, EA, MI
Navy - AS, MC, OS, SH
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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